

Modeling Stellar Energy Transport

Computational Astrophysics Project Report

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1 Introduction

A star is in Thermal equilibrium, so the rate at which energy is generated by nuclear fusion in the core is equal to the rate at which energy is radiated away (luminosity). This balance makes the star stable. Depending on the local temperature gradient, the energy transport occurs either by radiative or convective method. The goal of this project is to build a model of a star that determines the radial profiles of P , T , ρ , L and checks if transport is radiative or convective. The model will integrate inward from the surface towards core using the stellar structure equations and predicts the interior structure of a star.

This model must satisfy these goals:

- values of L , m , r should approach zero or at least within 5% of L_0 , M_0 , r_0
- core must extend to at least 10% of the solar radius
- convection zone must be present near the surface spanning at least 15% of the solar radius.

2 Use of this project

Such model can simulate different types of stars. A stellar structure model provides the link between these observables and the otherwise inaccessible interior, predicting the central temperature, density, pressure, and the locations of convective and radiative zones only based on its surface properties like temperature, radius, luminosity etc.

3 Stellar Structure Equations

The structure of the stellar interior is governed by four ODEs:

$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho} \tag{1}$$

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} \tag{2}$$

$$\frac{\partial L}{\partial m} = \varepsilon \tag{3}$$

$$\frac{\partial T}{\partial m} = \begin{cases} \nabla^* \frac{T}{P} \frac{\partial P}{\partial m} & \text{if } \nabla_{\text{stable}} > \nabla_{\text{ad}} \\ -\frac{3\kappa L}{256\pi^2 \sigma r^4 T^3} & \text{otherwise} \end{cases} \tag{4}$$

where m is mass coordinate, κ is the opacity, ε is the nuclear energy generation rate. The condition in Equation (4) determines if the local energy transport is radiative or convective at each mass shell.

4 Equation of State

The total pressure includes both radiation pressure and ideal gas pressure:

$$P = \frac{4\sigma}{3c}T^4 + \frac{\rho k_B T}{\mu m_u} \quad (5)$$

where c is the speed of light, k_B is Boltzmann's constant, m_u is the atomic mass unit, and μ is the mean molecular weight defined as:

$$\mu = \left(2X + \frac{3Y}{4} + \frac{Z}{2}\right)^{-1} \quad (6)$$

with solar composition $X = 0.7$, $Y = 0.29$, $Z = 1 - X - Y$. This gives $\mu \approx 0.618$. The density is:

$$\rho = \left(P - \frac{4\sigma T^4}{3c}\right) \frac{\mu m_u}{k_B T} \quad (7)$$

5 Methods

5.1 Opacity and Energy Generation Tables

We had been given 2 tables called `opacity.txt` and `epsilon.txt`. I first skipped comment lines and extracted values, with each row had $\log_{10} R$ values and the first column was $\log_{10} T$ values.

I did linear interpolation using `scipy.interpolate.RegularGridInterpolator`. If the given T or R fall in between given values, a linear interpolation is performed between the nearest two values to get estimate of κ and ϵ values. When T or R fall outside the table bounds, I set `bounds_error=False`, `fill_value=None`, which extrapolates linearly and gives a warning.

Units are also converted at the output: κ is returned in SI units ($\text{m}^2 \text{kg}^{-1}$) via the conversion $1 \text{ cm}^2 \text{g}^{-1} = 0.1 \text{ m}^2 \text{kg}^{-1}$, and ε in $\text{J kg}^{-1} \text{s}^{-1}$ via $1 \text{ erg g}^{-1} \text{s}^{-1} = 10^{-4} \text{ J kg}^{-1} \text{s}^{-1}$.

5.2 Convective Stability and the Temperature Gradient

Then I evaluated if the local temperature gradient would be radiative or convective at each integration step. Convection happens if $\nabla_{\text{stable}} > \nabla_{\text{ad}}$, where:

$$\nabla_{\text{stable}} = \frac{3\kappa\rho H_P L}{64\pi r^2 \sigma T^4}, \quad \nabla_{\text{ad}} = \frac{P}{T\rho C_P} \quad (8)$$

here $H_P = k_B T / (\mu m_u g)$ is the scale height, $g = Gm/r^2$ is gravitational acceleration, and $C_P = 5k_B / (2\mu m_u)$ is the specific heat at constant pressure for ideal gas. All these were calculated at each mass shell.

Also, we have to define 2 cases:

$$\nabla_{\text{conv}} = \begin{cases} \nabla_{\text{stable}} & \text{if } \nabla_{\text{stable}} < \nabla_{\text{ad}} \\ \nabla^* & \text{if } \nabla_{\text{stable}} > \nabla_{\text{ad}} \end{cases} \quad \begin{array}{l} \text{i.e., convection doesn't happen} \\ \text{i.e., convection happens (convectively unstable)} \end{array} \quad (9)$$

For this we find real root ξ of the cubic equation:

$$\xi^3 + \frac{U}{l_m^2}\xi^2 + \frac{U^2\Omega}{l_m^3}\xi + \frac{U}{l_m^2}(\nabla_{\text{ad}} - \nabla_{\text{stable}}) = 0 \quad (10)$$

where $U = \frac{64\sigma T^3}{3\kappa\rho^2 C_P} \sqrt{H_P/g}$, $l_m = H_P$, and $\Omega = 4/l_m$. The cubic is solved using `numpy.roots`. The cubic equation for ξ can have up to three positive roots, the smallest positive real root is selected, and ∇^* is then:

$$\nabla^* = \xi^2 + \frac{U\Omega}{l_m}\xi + \nabla_{\text{ad}} \quad (11)$$

If all roots are negative or complex, then ∇^* cannot be calculated and we instead use the radiative gradient, ∇_{stable} (see Eq. 9).

This ∇^* is then used in the stellar equation 4

5.3 ODE Integration: Adaptive RK45

The four structure equations are integrated inward from the stellar surface ($m = M_0$) to a small central mass fraction ($m = 10^{-4} M_0$) using `scipy.integrate.solve_ivp` with the RK45 method and tolerances `rtol=1e-4`, `atol=1e-8`. The adaptive timestep control in RK45 automatically adjusts the step size in m to maintain accuracy. Also, RK4 is more accurate than Euler or RK2, because it computes four gradients. The program is terminated in 2 cases: near center when the radius r drops below a threshold (100 m); or if P , T , or r become negative, which means there is some problem in model or initial conditions.

Reasoning of using mass coordinate I use $t_{\text{span}} = (M_0, m_{\text{min}})$ in `solve_ivp` function. So that means solver starts with $t = M_0$ at surface and integrates until it reaches $t = m_{\text{min}}$ near core. The mass coordinate m is used as it decreases smoothly whereas radius r can vary by orders of magnitude.

Initial conditions. The integration begins with solar surface values: $L_0 = L_{\odot}$, $R_0 = R_{\odot}$, $M_0 = M_{\odot}$, $T_0 = 5770$ K, and $\rho_0 = 1.42 \times 10^{-7} \bar{\rho}_{\odot}$, where $\bar{\rho}_{\odot} = 1.408 \times 10^3 \text{ kg m}^{-3}$. The initial pressure P_0 is derived directly from the equation of state at (T_0, ρ_0) .

6 Results

6.1 Sanity Checks

The sanity checks printed the numerically computed value, and the relative error for each test point, and issues warning if any error exceeds 5%. But at the end all opacity values were within 5% of the expected values (maximum relative error < 5%), and the two epsilon reference points were also within tolerance. This is shown here:

PASSED: opacity max rel error = $2.706 \times 10^{-2} \leq 5\%$

PASSED: epsilon max rel error = $1.24 \times 10^{-2} \leq 5\%$

6.2 Best-Fit Model

I tried running directly using solar values, but it did not satisfy all three goals simultaneously (central m , r , L going to zero; core extending to $\geq 10\%$ of R_0 ; convective envelope spanning $\geq 15\%$ of R_0).

After varying ρ_0 by factors from 1 to 1000, L_0 from 0.5 to $1 L_{\odot}$, and R_0 from 0.8 to $1.2 R_{\odot}$ — a model with $\rho_0 = 100 \rho_{0,\odot}$ and $L_0 = 0.8 L_{\odot}$ produced the best agreement with structural

goals. To do this I just changed values in wrapper function `run_model_with_factors`, in which all parameters were in multiples of solar values. The key results for this model are:

- $m_c/M_0 = 10^{-4}$ (central mass fraction, effectively zero)
- $r_c/R_0 \approx 5.84 \times 10^{-2}$ (central radius fraction)
- $L_c/L_0 \approx 0.767$ (central luminosity fraction)
- Core radius (defined as the largest r where $L < 0.995 L_0$): $= 0.236 R_0$ ($\geq 10\%$ goal met)
- Convective envelope width near surface: $= 0.205 R_0$ ($\geq 15\%$ goal met)

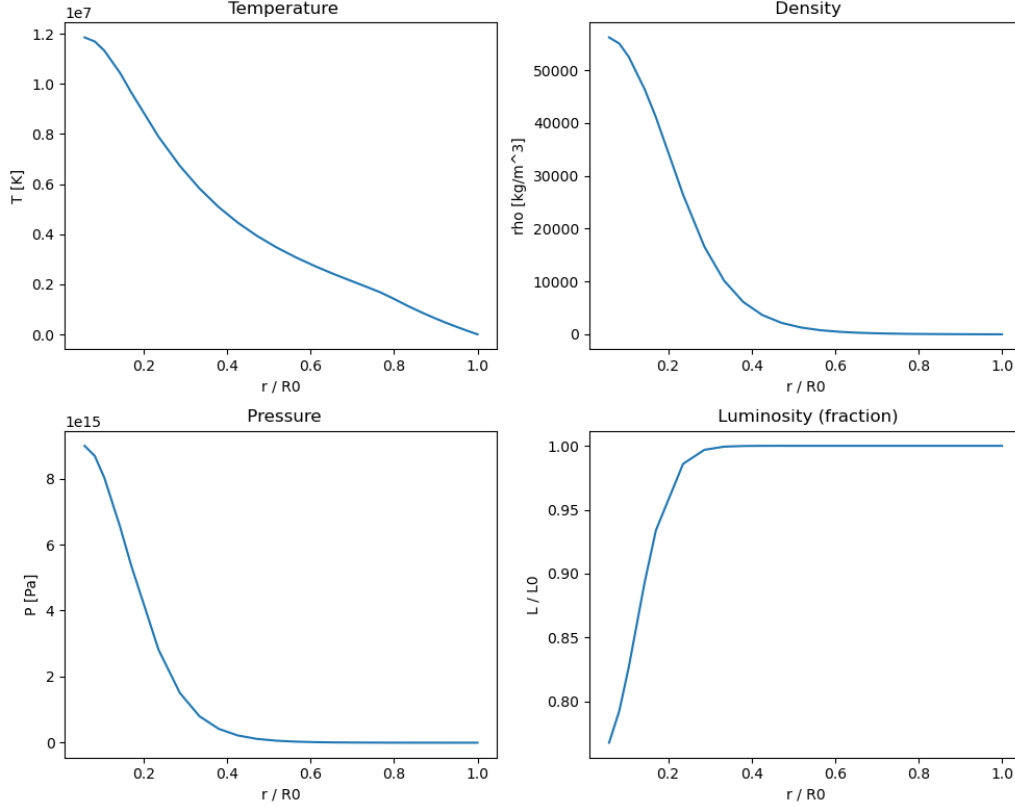


Figure 1: Radial profiles of temperature $T(r)$, density $\rho(r)$, pressure $P(r)$, and normalised luminosity L/L_0 for the best-fit model ($\rho_0 = 100 \rho_{0,\odot}$, $L_0 = 0.8 L_\odot$). The temperature rises from 5770 K at the surface to $\sim 12 \times 10^6$ K at the centre, similar to standard solar models (central temperature $\sim 1.57 \times 10^7$ K).

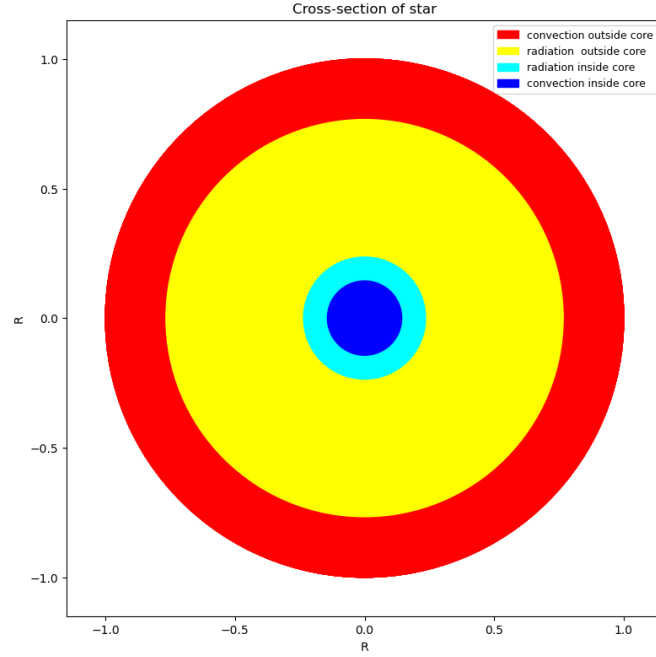
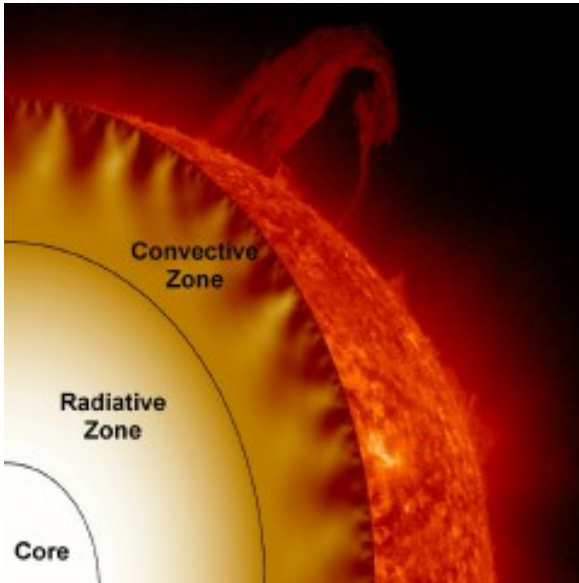


Figure 2: Shows the cross-section of the star. The convective envelope is clearly visible as a thick outer shell, while the interior is radiative. The core also has convection in inner part. This is similar to the solar interior (see diagram by NASA [3a](#): a radiative zone from the centre to $\sim 0.7 R_{\odot}$ and a convective envelope in the outer 30%)



(a) [NASA](#)

Interior Zones of the Sun				
Zone	R/R(0)	Temperature (K)	Density (g/cm ³)	Energy Transport
Core	0.0 - 0.25	~ 15,000,000 - 8,000,000	~ 160 - 10	Radiative
Radiative	~ 0.25 - 0.85	~ 8,000,000 - 500,000	~ 10 - 0.01	Radiative
Convective	~ 0.85 - 1.00	~ 500,000 - 10,000	< 0.01	Convective

(b) This is a table I found from [University of Rochester](#)

Figure 3: To compare our results with actual solar structure. We see that except the convective part in core, this is the exact structure we got in Fig 2

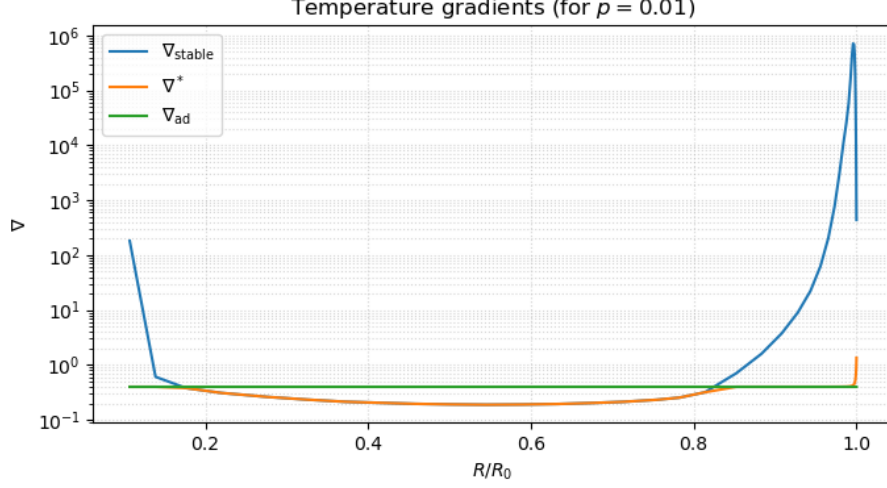


Figure 4: Shows the temperature gradients ∇_{stable} , ∇_{ad} , and ∇^* as a function of $r/R_\odot$. Near the surface ($r/R_0 \geq 0.7$) and at the inner core ($r/R_0 \leq 0.2$), $\nabla_{\text{stable}} > \nabla_{\text{ad}}$, triggering convection. The actual convective gradient ∇^* lies between ∇_{ad} and ∇_{stable} , consistent with the expectation that $\nabla_{\text{ad}} \leq \nabla^* \leq \nabla_{\text{stable}}$.

6.3 Challenges Encountered

R values out of bound In the deepest layers a lot of shells exceeded the opacity table limits for $\log R$. I got warnings like `WARNING [opacity]: log10(R)=1.162 outside [-8.0, 1.0]. Extrapolating.` The maximum such value I got was $\log R = 2.017$. So I think model reached to higher density than the table was designed for, maybe in the deep interior. So the code extrapolates opacities there, this may add some uncertainty in the central values

Luminosity At the end I got $L_c/L_0 = 0.768$. Means at the innermost point, luminosity is still 77% of L_0 when the goal was that it should be close to 0. Maybe integration is stopping way too early before the luminosity can go down. Or maybe the initial L_0 is too high for given value of nuclear energy generated in core by fusion (ε). But the other parameters are close to zero, so I do not understand why this happend only with luminosity.

Core convection In Fig 3b, the core is said to be fully radiative. But I got both radiative and convective zones inside the core (see Fig 4, 2). This again might be due to the fact that L_c was too high as compared to expected.

7 Conclusion

A one-dimensional stellar structure model for a Sun-like star was implemented successfully. The model nearly satisfies all three structural goals (except Luminosity = 0 part). The temperature and density profiles are nearly consistent with solar models, with a central temperature of $\sim 1.2 \times 10^7$ K and central density of $\sim 5 \times 10^4$ kg m $^{-3}$. Despite all simplifications, Convection is seen near the surface as expected, and radiative region is seen from core upto the convection region.